Shlok Patel

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U.S. Citizen | *Boston, MA — Available Jan – Sep 2027*

Education

Northeastern University, Boston, MA — May 2029
Candidate for B.S. in Computer Science and Business Administration (Financial Technology) — ***GPA: 3.74/4.00***
Coursework: Software Engineering, Object-Oriented Design, Algorithms & Data Structures, Computer Systems, Databases, Corporate Finance, Business Statistics, Financial Accounting, Information Systems & Digital Technologies
Honors & Societies: Dean's List, Honors College, NU ACES, NU Disrupt, Teaching Assistant (OOD)

Technical Skills

Languages: Python, Java, SQL, C++, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Flask, FastAPI, PyTorch, Pandas, NumPy, SciPy, scikit-learn
AI & ML: Claude Code, GitHub Copilot, LLM APIs (Groq, Gemini), model evaluation, agentic coding workflows
Developer Tools & Databases: MongoDB, PostgreSQL, AWS, Supabase, Prisma, Docker, Git, CI/CD, Linear

Experience

Boston Bruins — Sep 2026 – Present
Data Analyst (Professional Practicum DS 4535) — *Boston, MA*

- Formulate a reproducible **Python** data pipeline (**pandas, scikit-learn**) for a predictive-modeling engagement.
- Spearhead requirements elicitation with stakeholders, scoping objectives into a delivery plan across **Agile** sprints.

NU Intelligent Automation and IoT Research Group — Jul 2026 – Present
Undergraduate Research Assistant — *Boston, MA*

- Engineer an ML pipeline (**librosa, Praat, PyTorch**) fusing voice, tapping, and gait for Parkinson's detection.
- Architect gated and attention fusion nets plus a stacking meta-learner, benchmarking **8+** models leakage-free.
- Elevate **AUROC** from **0.720 to 0.774** (p=0.080) across **2,685** subjects, hardened by **100+** unit tests.

Student Government Association — Jan 2026 – Present
Lead Software Engineer — *Boston, MA*

- Streamline a production CMS serving **5,000+** students, cutting content publish time **10%** for non-technical editors.
- Architect version history, rollback, and role-based access on **Next.js**/**Prisma**, keeping every change auditable.
- Engineer **Prisma**-validated endpoints enforcing schema integrity, eliminating malformed writes to **PostgreSQL**.
- Manage the **SDLC** for a team of **5+** engineers as one of **3 Tech Leads**: tickets, sprint planning, code review.

Projects

NUHire | *Next.js, Express, Socket.IO, MySQL, Keycloak, Docker* — Sep 2026 – Present

- Build a **10-check** GitHub Actions CI/CD gate (tests, Docker, migrations, security audit) gating all merges to **main**.
- Execute end-to-end QA of the live classroom flow, logging **60+** issues including a critical mid-session lockout bug.
- Automate **dbmate** migrations to fail-stop before startup and fix the Docker build to run cross-arch (**x86/ARM**).

PhishAware | *Next.js, TypeScript, MongoDB, FastAPI, Groq/Gemini* — Jul 2026 – Sep 2026

- Led a 3-person team shipping a full-stack phishing-awareness platform across **6** attack vectors.
- Migrated the data layer across **44 files** and built a multi-tenant org backend with role-based access control.
- Generated a contract-first **OpenAPI** pipeline (**57 operations**, **45 paths**) producing Zod and React Query hooks.
- Integrated from-scratch **OIDC**/**PKCE** SSO and a **Groq**/**Gemini** LLM scenario pipeline with provider failover.

Derivative Hedging via Reinforcement Learning | *Python, PyTorch, Stable-Baselines3, NumPy* — Jan 2026 – Mar 2026

- Engineered a reproducible harness (**6** strategies, **4** regimes, **10K**-resample paired bootstrap) for hedging policies.
- Validated the RL-vs-delta edge insignificant (p=0.29–0.97); **Whalley-Wilmott** beat it with **80%** fewer trades.
- Redesigned the reward as a **CVaR** objective, tracing via drift-ablation how a directional tilt gamed it.

Interests & Leadership

Interests: Community-based volunteering, Chess, Graphic Design, Hiking, Cooking, American Football
Leadership: Project-Series Mentor, NU Oasis — lead groups of 5+ on their first-ever software development journeys.